Image recreation using Genetic Algorithms and Color metrics

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Description

This study uses a GA to evolve randomly generated triangular shapes, optimizing their color (via CIE Lab $\Delta E*94$) and structural alignment with an test image. The method balances exploration of geometric configurations with exploitation of high-fidelity solutions, demonstrating the potential of evolutionary computation to computationally approximate artistic masterpieces.

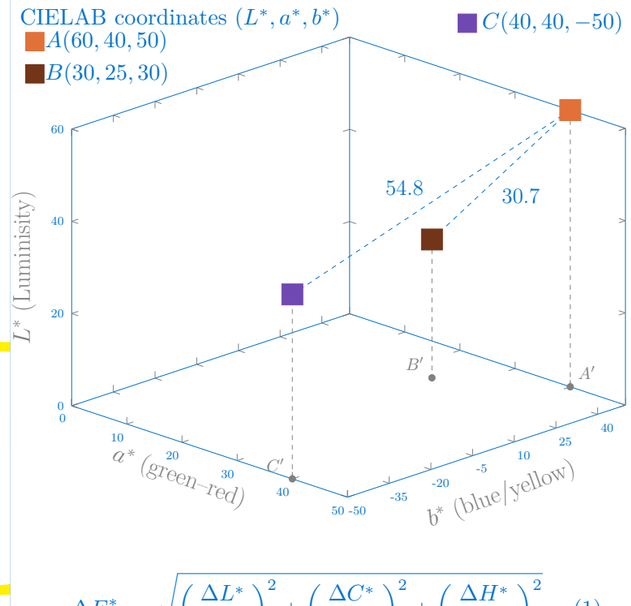
Methodology

1. Each **individual** represents an image composed of triangles.

Example of a first individual.

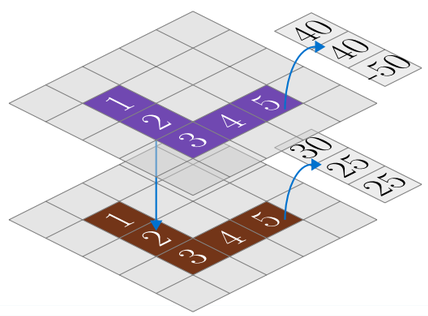
2. Each image is evaluated using the CIE Lab $\Delta CIE1994$ metric to measure **color similarity** as the mean value. Equation (**??**) is used in each pixel.
3. The GA selects (tournament), crosses, and mutates the best solutions to improve the approximation.
4. The process is repeated over several generations until an optimal solution is reached.
5. A stop condition is applied when fitness is < 0.01 for 3 consecutive individuals.

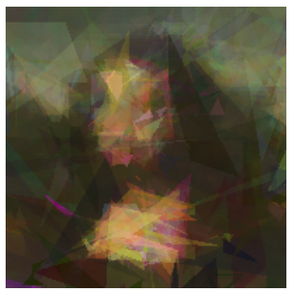
Results



$$\Delta E^*_{94} = \sqrt{\left(\frac{\Delta L^*}{k_L S_L}\right)^2 + \left(\frac{\Delta C^*}{k_C S_C}\right)^2 + \left(\frac{\Delta H^*}{k_H S_H}\right)^2} \quad (1)$$

Each individual is compared with the test image comparing pixel by pixel extracting the $L^*a^*b^*$ coordinates:

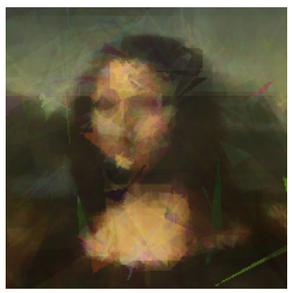




Triangles capture edges.



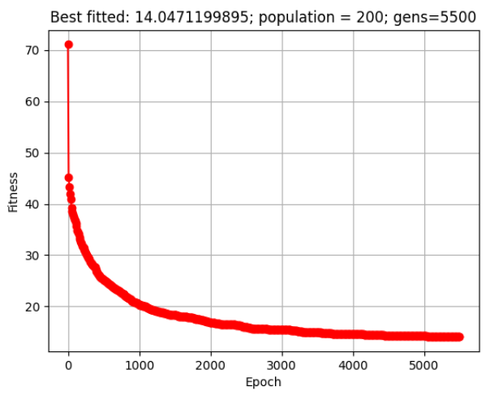
Operators improves details.



Captures background colors.



At 5,000 generations.



Discussion

1. The algorithm showed rapid convergence in early generations (≤ 50), followed by stagnation, indicating a trade-off between exploration and exploitation.
2. Triangular shapes captured edges effectively but limited color blending. While $\Delta E\ CIE1994$ correlated well with human perception, it overlooked structural texture.
3. Larger populations (≥ 100) enhanced diversity at higher computational costs.

Conclusion

1. The genetic algorithm successfully approximated the Mona Lisa using triangular shapes and the $\Delta CIE1994$ color metric, achieving a 68 % reduction in color discrepancy over 100 generations.
2. While computationally intensive, the method demonstrated the potential of evolutionary strategies in artistic replication.
3. Future implementations could benefit from hybrid approaches (e.g., neural-guided mutations) and texture-aware metrics to overcome current limitations in smoothness and structural fidelity.

References

[1] Tian, Y., Ha, D. (2022). Modern Evolution Strategies for Creativity: Fitting Concrete Images and Abstract Concepts. In: Artificial Intelligence in Music, Sound, Art and Design. EvoMUSART 2022. Lecture Notes in Computer Science, vol 13221. Springer, Cham.

[2] Melgosa, M. (2023). CIE94, History, Use, and Performance. In: Shamey, R. (eds) Encyclopedia of Color Science and Technology. Springer, Cham.

Acknowledgments

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alteredqualia.com/visualization/evolve/